Study of Neutron Star Properties with various hybrid models.

By

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DECLARATION

I, **Debabrata Dey**, hereby declare that the investigations presented in the thesis have been carried out by me. The matter embodied in the thesis is original and has not been submitted earlier as a whole or in part for a degree/diploma at this or any other Institution/University.

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List of Publications for the thesis

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   JCAP 01 (2025) 056, arXiv:2401.02190v2 [astro-ph].

Work in progress

1. *Core or Halo? Two-Fluid Analysis of Dark Matter-Admixed Quarkyonic Stars in the Multi-Messenger Era*
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2. *Quarkyonic Model for Neutron Star Matter: A Relativistic Mean-Field Approach*
   Ankit Kumar, **Debabrata Dey**, Shamim Haque, Ritam Mallick, S. K. Patra
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1. *Dark Matter Effects on f-mode Oscillations in Quarkyonic Stars*
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3. Quarkyonic model for dark matter admixed neutron Star: A RMF perspective
   Debabrata Dey, H. C. Das, Ankit Kumar, Jeet Amrit Pattanaik, R.N. Panda, S. K. Patra
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4. *Influence of Dark Matter on Neutron Star Curvature within the Quarkyonic Model: A Relativistic Mean Field Approach*
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5. Tidal deformability of dark matter admixed neutron star within quarkyonic model in the conjunction of relativistic mean field formalism
   Debabrata Dey, H. C. Das, Ankit Kumar, Jeet Amrit Pattanaik, R.N. Panda, S. K. Patra
   DAE Symp.Nucl.Phys. 67 (2024) 805-806.

6. f-Mode Universal Relation of Dark Matter Admixed Quarkyonic Stars
   Debabrata Dey, Jeet Amrit Pattanaik, M. Bhuyan, R.N. Panda, S. K. Patra
   Springer Proc.Phys. 432 (2026) 231-234 .

7. Quarkyonic model for dark matter admixed neutron Star: A RMF perspective
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DEDICATIONS

Dedicated To Maa, Bapi and Bhai

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(Debabrata Dey)

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CHAPTER 1

Introduction

1.1 Strong Interaction

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1.4 Conventions

- Greek indices run from 0 to 3 and denote space–time components (with 0 being the time component), Latin indices run from 1 to 3 and denote spatial components. We use the Einstein sum convention for these indices.
- Latin indices in roman typeset, e.g. $\mathrm{n}, \mathrm{p}, \mathrm{x}, \ldots$, are constituent indices. We place them liberally upstairs or downstairs in order to avoid cluttering and they do not obey any kind of summation convention.
- The signature of the metric is $(-, +, +, +)$ which means that time-like vectors have a negative length.

- Unless stated otherwise, we carry out our calculations in units in which $G = c = M_\odot = 1$. Whenever appropriate, we quote physical quantities in cgs-units; occasionally we use MeV.

- Covariant derivatives of an arbitrary quantity f with respect to x^μ are written as $\nabla_\mu f$, and we occasionally use the abbreviations $\partial_\mu f$ or $f_{,\mu}$ for partial derivatives.

- We abbreviate radial derivatives with a prime, $f' = \partial_r f$, and temporal derivatives with a dot, $\dot{f} = \partial_t f$.

- For brevity, we will occasionally make use of the following short-hand notation regarding tensor indices: for any tensor $t_{\mu\nu}$, round brackets denote symmetrisation,

  $$2t_{(\mu\nu)} := t_{\mu\nu} + t_{\nu\mu},$$

  whereas square brackets are used to denote anti-symmetrisation,

  $$2t_{[\mu\nu]} := t_{\mu\nu} - t_{\nu\mu}.$$

We plan the thesis in the following fashion. In Chapter 2, we introduce the quantum mechanical approach for constructing the formalism of neutrino oscillations. We provide fundamental insights into two-flavor and three-flavor neutrino oscillations, both in vacuum and Earth's matter. Furthermore, we conduct an in-depth exploration of the degeneracies associated with the neutrino oscillation parameters and highlight the prevailing tensions among various neutrino oscillation experiments. Chapter 3 presents a comprehensive overview of the evolution of long-baseline experiments, tracing their trajectory from inception to their current advancements. In Chapter 4, we illustrate how the contemporary and next-generation long-baseline (LBL) neutrino experiments hold immense potential in probing the deviation of the atmospheric mixing angle (θ_{23}) from maximal mixing, resolving the octant ambiguity ($\theta_{23} > 45^\circ$ or $< 45^\circ$), and significantly enhancing the precision of the oscillation parameters θ_{23} and Δm^2_{31}. We particularly emphasize the role of the upcoming Deep Underground Neutrino Experiment (DUNE) in surpassing present measurements within the three-neutrino framework. Chapter 5 delves into the synergistic interplay between DUNE and T2HK (Tokai-to-Hyper-Kamiokande) in resolving degeneracies among neutrino oscillation parameters. We investigate how the combined sensitivity of these experiments strengthens key aspects of neutrino oscillation phenomenology, including the establishment of non-maximal θ_{23}, exclusion of the incorrect octant, and precision measurements of the 2-3 oscillation parameters of the PMNS matrix. These pursuits are particularly compelling as neutrino oscillation research transitions into the precision era. Additionally, we demonstrate how the complementary features of DUNE and T2HK facilitate an enhanced physics reach, achieving substantial sensitivity at lower exposures compared to their individual baseline configurations. Lastly, in Chapter 6, we explore how the next-generation LBL experiments can uncover signatures of the flavor-dependent long-range neutrino interactions within the neutrino sector. Specifically, we analyze how these experiments can probe the presence of the gauged $(L_e - L_\mu)$, $(L_e - L_\tau)$, and $(L_\mu - L_\tau)$ symmetries through the aforementioned oscillation phenomenology, thereby establishing compelling evidence for such interactions with a high degree of confidence. Finally, Chapter 7 ends this thesis with some insightful concluding remarks, encapsulating the key takeaways, and their importance in the neutrino oscillation frontier.

CHAPTER 2

Theoretical framework

Neutrino oscillation is one of the pioneering discoveries in the history of particle physics. It has opened a new horizon for the community to utilize this second most abundant particle to reveal several mysteries of the Universe [1]. The cornerstone of the neutrino oscillation was first laid in the Super-Kamiokande detector (Japan) in 1998 [2] by providing experimental evidence in their data. Non-zero neutrino mass is the first experimental proof of BSM physics, for which the Nobel Prize was awarded to Takaaki Kajita and Arthur McDonald in 2015 [3, 4].

Before entering into the formalism of neutrino oscillation, it is essential to know why neutrinos show the flavor transition (named "neutrino oscillation") [5, 6], whereas their charged counterparts do not show oscillations. Neutrino oscillation is a mass-induced phenomenon. Due to the non-zero value of neutrino mass and the non-degenerate values of the mass eigenstates [7], the flavor eigenstates and mass eigenstates of neutrinos are quite different. The tiny mass-squared differences between the neutrino mass eigenstates ($\sim 10^{-5}$ to $10^{-3}\,\mathrm{eV}^2$) protect the coherence of the superposition [8, 9] of mass eigenstates to the flavor states. In contrast, the mass differences between the leptons are so huge ($\sim$ 100 MeV to GeV scale) that any superposition of the mass eigenstates of the leptons decoheres almost immediately before their significant propagation.

This chapter is planned in the following way. Section 2.1 outlines the fundamental theoretical framework of neutrino oscillations in the vacuum, examining the formalism in the two-flavor framework. In section 2.4.2, the extension of this formalism in the three-flavor landscape is described. In section 2.5, we introduce matter-effect in our discussion, incorporating standard neutrino-matter interactions within both two-flavor and three-flavor scenarios. Section **??** provides an overview of several relevant neutrino oscillation experiments and their contributions to measuring oscillation parameters. In section **??**, we present the current status of the six oscillation parameters and the so-called eight-fold degeneracies, discussing their experimental determination. Finally, in section **??**, we provide a concluding overview of this chapter.

2.1 Neutron star Matter

There are several approaches to construct the neutrino oscillation framework. Neutrinos are fermions and occupy their places in the lepton family of the SM, in the form of $SU(2)$ doublets. So, ideally, the Dirac equation should lead us to the neutrino oscillation probability. However, neutrino oscillation does not manifest any spin nature of itself, and hence, for the ease of our calculations, we use the Schrödinger equation considering neutrino as a free particle instead of assuming it as a wave packet. Neutrinos are produced or detected at their flavor eigenstates (ν_e, ν_μ, or ν_τ) or weak eigenstates (or gauge states), but they are in the mass eigenstates or physical states (ν_1 and ν_2) while propagating through the vacuum or the matter. The mass eigenstates and flavor eigenstates are considered to form a complete orthonormal basis to conserve the flavor transition probability, and hence a flavor eigenstate, ν_α is connected to the mass eigenstate ν_k through a unitary matrix U in the following fashion.

2.2 Quarkyonic Star Matter

2.3 Dark Matter

$$|\nu_\alpha\rangle = \sum_{k=1}^{2} U_{\alpha k} |\nu_k\rangle \,, \tag{2.1}$$

where α is the flavor index [$\alpha = \{e, \mu\}$, or $\{e, \tau\}$, or $\{\mu, \tau\}$] and k is the mass index, $U_{\alpha k}$ are the elements of the unitary matrix connecting the bridge between this two orthonormal basis, ensuring the probability conservation. The completeness of the basis indicates that $\sum_\alpha |\nu_\alpha\rangle \langle\nu_\alpha| = \sum_{k=1,2} |\nu_k\rangle \langle\nu_k| = \mathbb{1}$, whereas the orthonormality ensures that $\langle\nu_\alpha|\nu_\beta\rangle = \delta_{\alpha\beta}$ and $\langle\nu_i|\nu_j\rangle = \delta_{ij}$. The crucial point is to be noted here that the three mass eigenstates are non-degenerate (*i.e.*, the mass of each eigenstate must not be equal, $m_1 \neq m_2$, where m_k is the mass of the eigenstate ν_k). These mass eigenstates evolve with time while neutrinos propagate through the vacuum or the matter, and hence, the flavor eigenstates are also affected by time evolution. As the mass eigenstates are the eigenstates of the Hamiltonian, we can write the time evolution of the mass eigenstates in the following way.

Let us consider a system where a neutrino beam of flavor state $|\nu_\alpha(0,0)\rangle$ is generated at space-time point $(x,t) = (0,0)$.
We know while the neutrino will propagate through the vacuum they are in mass eigenstates and its time evolution will be governed by the time-dependent Schrödinger equation, *i.e.*,

$$i\frac{\partial}{\partial t} |\nu_k(x,t)\rangle = E\,|\nu_k(x,t)\rangle \tag{2.2}$$

$$= \frac{-1}{2m_k}\frac{\partial^2}{\partial^2 x} |\nu_i(x,t)\rangle \,, \tag{2.3}$$

where, $k = 1, 2$. Here, we have used the natural units (*i.e.*, $\hbar$=1).

Now,

$$i\frac{\partial}{\partial t}\left|\nu_k(x,t)\right\rangle = E\left|\nu_k(x,t)\right\rangle \tag{2.4}$$

$$\Rightarrow\frac{\partial\left|\nu_k(x,t)\right\rangle}{\left|\nu_k(x,t)\right\rangle} = -iE\partial t. \tag{2.5}$$

Performing partial integration, we get

$$\left|\nu_k(x,t)\right\rangle = e^{-iE_k t}\left|\nu_i(0,0)\right\rangle, \tag{2.6}$$

and the trial solution $\left|\nu_k(x,t)\right\rangle = e^{ip_k x}$ satisfies the equation

$$\frac{-1}{2m_k}\frac{\partial^2}{\partial^2 x}\left|\nu_k(x,t)\right\rangle = E\left|\nu_k(x,t)\right\rangle. \tag{2.7}$$

So finally,

$$\left|\nu_k(x,t)\right\rangle = e^{-i(E_k t - p_k x)}\left|\nu_k(0,0)\right\rangle \tag{2.8}$$

$$= e^{-i\phi_k}\left|\nu_k(0,0)\right\rangle, \tag{2.9}$$

where,

$$\phi_k = E_k t - p_k x. \tag{2.10}$$

At some later space-time point (x,t), the time evolved flavor state $\left|\nu\right\rangle_\alpha$ will be

$$\left|\nu_\alpha(x,t)\right\rangle = \sum_{k=1}^{2} U_{\alpha k}\left|\nu_k(x,t)\right\rangle \tag{2.11}$$

$$= \sum_{k=1}^{2} U_{\alpha k} e^{-i\phi_k}\left|\nu_k(0,0)\right\rangle. \tag{2.12}$$

So,

$$|\nu_k(0,0)\rangle = \sum_{\gamma} U^*_{\gamma k} |\nu_\gamma(0,0)\rangle \,. \tag{2.13}$$

Here γ is a dummy index of the unitary matrix while being inverted.
We have to find the evolution of flavor states, but from the Schrödinger equation, we got the time evolution of mass eigenstates as they are the eigenstates of the governing Hamiltonian.
So, substituting equation 2.13 into equation 2.12, we get,

$$|\nu_\alpha(x,t)\rangle = \sum_{k=1}^{2} U_{\alpha k} e^{-i\phi_k} \sum_{\gamma=1}^{2} U^*_{\gamma k} |\nu_\gamma(0,0)\rangle \tag{2.14}$$

$$= \sum_{\gamma} \sum_{k} U^*_{\gamma k} e^{-i\phi_k} U_{\alpha k} |\nu_\gamma(0,0)\rangle \,. \tag{2.15}$$

Transition amplitude for detecting a neutrino of flavor β at space-time point (t,x) when generation of a neutrino of flavor α at space-time point (0,0) is

$$\begin{aligned} A(|\nu_\alpha(0,0)\rangle \rightarrow |\nu_\beta(x,t)\rangle) &= \langle \nu_\beta(x,t)|\nu_\alpha(0,0)\rangle \\ &= \sum_{\gamma} \sum_{k} U_{\gamma k} e^{i\phi_k} U^*_{\beta k} \langle \nu_\gamma(0,0)|\nu_\alpha(0,0)\rangle \\ &= \sum_{\gamma} \sum_{k} U_{\gamma k} e^{i\phi_k} U^*_{\beta k} \delta_{\gamma\alpha} \\ &= \sum_{k} U_{\alpha k} e^{i\phi_k} U^*_{\beta k}. \end{aligned} \tag{2.16}$$

So, the oscillation probability is,

$$\begin{aligned} P(\nu_\beta \rightarrow \nu_\alpha) &= |A(|\nu_\beta(0,0)\rangle \rightarrow |\nu_\alpha(x,t)\rangle)|^2 \\ &= |\sum_{k} U_{\alpha k} e^{i\phi_k} U^*_{\beta k}|^2 \\ &= \sum_{k} U_{\alpha k} e^{i\phi_k} U^*_{\beta k} \sum_{j} U^*_{\alpha j} e^{-i\phi_j} U_{\beta j} \end{aligned}$$

$$= \sum_{j=1}^{2}\sum_{k=1}^{2} U_{\alpha k} U^{*}_{\beta k} U^{*}_{\alpha_j} U_{\beta j} e^{-i(\phi_j - \phi_k)}. \tag{2.17}$$

For the two-flavor oscillation, the (2×2) unitary matrix is just a rotation matrix in 2d.

$$U = \begin{pmatrix} U_{\alpha_1} & U_{\alpha_2} \\ U_{\beta_1} & U_{\beta_2} \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}, \tag{2.18}$$

where θ is a parameter called the mixing angle. For the two-flavor oscillation, it is the only mixing parameter as a 2×2 unitary matrix can be parameterized by only one parameter. We show the calculation of equation 2.17 term by term.

For $\underline{k = 1,\ j = 1;}$

$$\begin{aligned} P_1(\nu_\beta \to \nu_\alpha) &= U_{\alpha 1} U^{*}_{\beta 1} U^{*}_{\alpha_1} U_{\beta 1} e^{-i(\phi_1 - \phi_1)} \\ &= |U_{\beta 1}|^2 |U_{\alpha_1}|^2. \end{aligned} \tag{2.19}$$

For $\underline{k = 2,\ j = 2;}$

$$P_2(\nu_\beta \to \nu_\alpha) = |U_{\beta 2}|^2 |U_{\alpha_2}|^2. \tag{2.20}$$

For $\underline{k = 1,\ j = 2;}$

$$P_3(\nu_\beta \to \nu_\alpha) = U_{\alpha 1} U^{*}_{\beta 1} U^{*}_{\alpha 2} U_{\beta 2} e^{-i(\phi_2 - \phi_1)}. \tag{2.21}$$

For $\underline{k = 2,\ j = 1;}$

$$P_4(\nu_\beta \to \nu_\alpha) = U_{\alpha 2} U^{*}_{\beta 2} U^{*}_{\alpha 1} U_{\beta 1} e^{-i(\phi_1 - \phi_2)}. \tag{2.22}$$

Finally,

$$
\begin{aligned}
P(\nu_\beta \to \nu_\alpha) &= |U_{\beta 1}|^2 |U_{\alpha 1}|^2 + |U_{\beta 2}|^2 |U_{\alpha 2}|^2 + U_{\alpha 1} U_{\beta 1}^* U_{\alpha 2} U_{\beta 2}^* \left(e^{i(\phi_2 - \phi_1)} + e^{-i(\phi_2 - \phi_1)}\right) \\
&= (\sin^2\theta \cos^2\theta + \cos^2\theta \sin^2\theta) + 2\, U_{\alpha 1} U_{\beta 1}^* U_{\alpha 2} U_{\beta 2}^* \cos(\phi_2 - \phi_1) \\
&= 2\sin^2\theta \cos^2\theta + 2\cos\theta(-\sin\theta)\sin\theta\cos\theta\cos(\phi_2 - \phi_1) \\
&= 2\sin^2\theta\cos^2\theta\,[1 - \cos(\phi_2 - \phi_1)] \\
&= 4\sin^2\theta\cos^2\theta\sin^2\left(\frac{\phi_2 - \phi_1}{2}\right) \\
&= \sin^2(2\theta)\sin^2\left(\frac{\phi_2 - \phi_1}{2}\right).
\end{aligned}
\tag{2.23}
$$

By equation 2.10,

$$\phi_k = E_k t - p_k x. \tag{2.24}$$

So,

$$\phi_2 - \phi_1 = (E_2 - E_1)t - (p_2 - p_1)x. \tag{2.25}$$

At this point, we make our first assumption that neutrinos are relativistic, so $t = x = L$. Here, L is the length of the baseline, and E is the true neutrino energy. As we are in natural unit scale, $c = 1$, and

$$
\begin{aligned}
p_i &= \sqrt{E_i^2 - m_i^2} \\
&= E_i\sqrt{1 - \frac{m_i^2}{E_i^2}} \\
&\approx E_i(1 - \frac{m_i^2}{2E_i^2}) \\
&= E_i - \frac{m_i^2}{2E_i},
\end{aligned}
\tag{2.26}
$$

as neutrino masses are very small compared to their energy.

By equation 2.26,

$$\phi_2 - \phi_1 = \left(\frac{m_2^2}{2E_2} - \frac{m_1^2}{2E_1} \right) L. \tag{2.27}$$

Our next assumption posits that each neutrino flavor state is associated with a well-defined momentum p, implying that all mass eigenstates ν_i (i= 1,2) possess an identical momentum. This premise is grounded in the notion that all mass eigenstates emanate from a common source and traverse coherently along the same trajectory. It is to be noted here that mass eigenstates are assumed as plane wave solutions. Henceforth, we will consider $E_1 = E_2 = E$ in our calculation.
Now,

$$\begin{aligned} \phi_2 - \phi_1 &= \left(\frac{m_2^2}{2E} - \frac{m_1^2}{2E} \right) L \\ &= \frac{\Delta m^2 \, L}{2E}, \end{aligned} \tag{2.28}$$

where, $\Delta m^2 = m_2^2 - m_1^2$.
By equation 2.23, we can write the probability of the neutrino oscillation from the flavor α to the flavor β is

$$P(\nu_\alpha \to \nu_\beta) = \sin^2 (2\theta) \sin^2 \left(\frac{\Delta m^2 \, L}{4E} \right). \tag{2.29}$$

Here, the first term signifies the amplitude part of the oscillation (governed by the parameter "mixing angle" θ), and the second term signifies the frequency part (governed by the parameter "mass-squared difference" Δm^2). They are fixed in Nature. Here, we have calculated the oscillation probability by imposing natural units (*i.e.*, $\hbar$=c=1).

Now, if we want this oscillation probability formula compatible with the data obtained from the experiments, we need to convert the units of the equation 2.29

$\left[\Delta m^2\,(\text{in eV}^2),\, L\,(\text{in eV}^{-1}),\, E\,(\text{in eV})\right]$ to the units used by the experiments [*i.e.*, $\Delta m^2\,(\text{in eV}^2)$, $L\,(\text{in km})$, $E\,(\text{in GeV})$]. The quantity $\left(\dfrac{\Delta m^2\,L}{4E}\right)$ is in the argument of a sinusoidal function and needs to be dimensionless. After implementing those conversion factors, the final form of the neutrino oscillation probability from a flavor α to be transited to a flavor β through vacuum in the two-flavor landscape is,

$$P(\nu_\alpha \to \nu_\beta) = \sin^2(2\theta)\sin^2\left(1.27 \times \frac{\Delta m^2\,L}{E}\right), \tag{2.30}$$

where L is the length over which neutrino changes its flavor (in km), E is the true neutrino energy (in GeV), and Δm^2_{21} is the mass-squared difference (in eV^2), where $\Delta m^2 = m_2^2 - m_1^2$. It is noteworthy that, two flavor neutrino oscillation only suggests the non-trivial mixing and mass-induced flavor conversion in the neutrino sector but cannot predict anything regarding the CP violation. Similarly, the expression of the survival (or disappearance) probability in the two-flavor landscape can be written as,

$$P(\nu_\alpha \to \nu_\alpha) = 1 - \sin^2(2\theta)\sin^2\left(1.27 \times \frac{\Delta m^2\,L}{E}\right). \tag{2.31}$$

Equation 2.30 resembles with the standard oscillation formula in the acoustic medium, where the first part signifies the amplitude part (governed by the mixing angle θ), whereas the second part depicts the frequency part (governed by the mass-squared difference Δm^2). If θ becomes $45°$, appearance probability (2.30) becomes maximum. Hence the value $\theta = 45°$ is called the maximal mixing (MM) value [10]. Any other values of θ show lesser appearance probability than the maximal mixing for a given set of values of oscillation parameters. On the other hand, the shift of the frequency can be tuned to different energy and baseline lengths by controlling the value of Δm^2.

Figure 2.1 illustrates that the variations in the mixing angle enable precise modulation of the amplitude of the neutrino oscillation probability, while adjustments to the mass-squared difference primarily govern the phase shift of the oscillation pattern. The mixing angle θ signifies the extent of flavor mixing, with maximal mixing corresponding to the highest transition probability or the deepest suppression in survival probability. Moreover, in the oscillation frequency domain, an increase in the value of

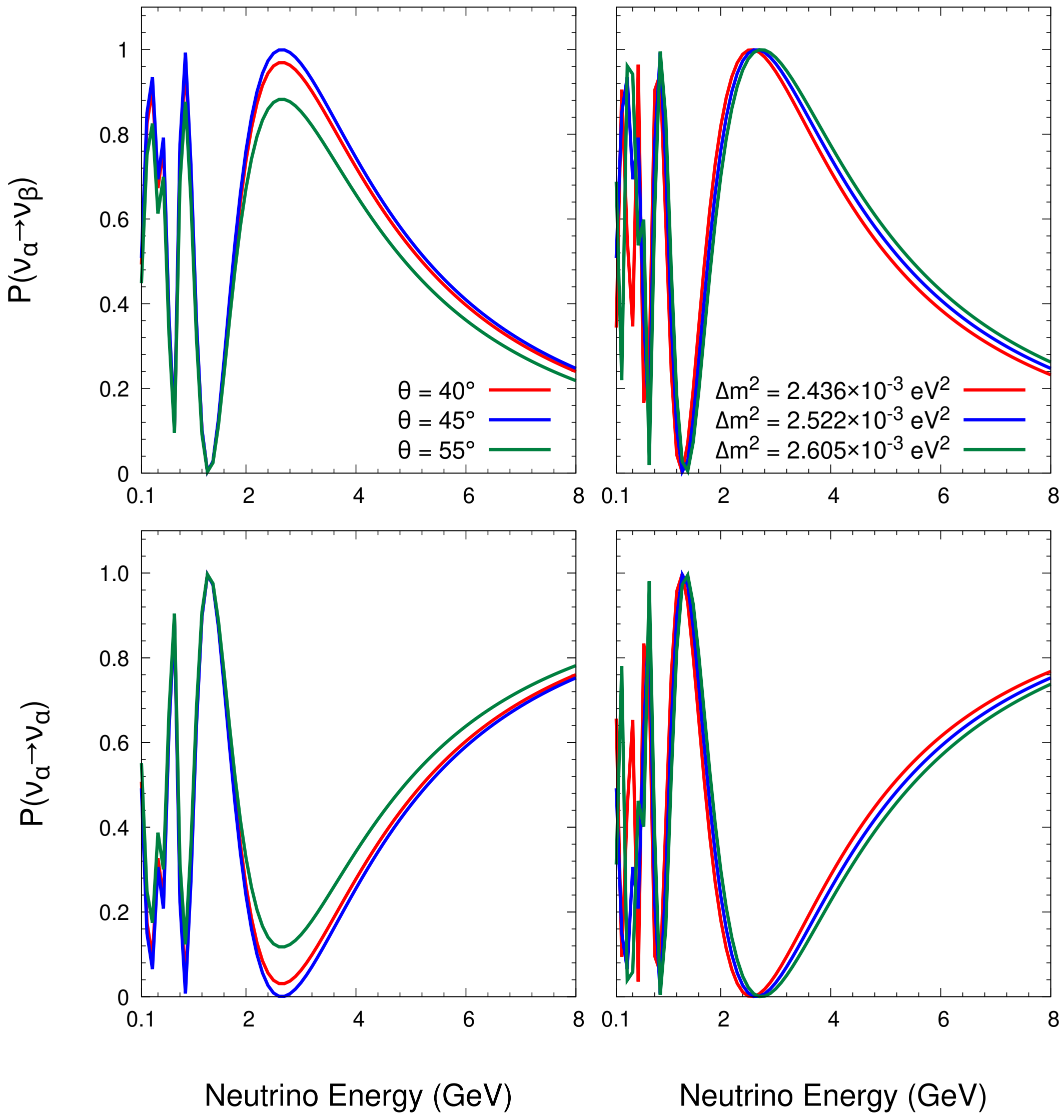

Figure 2.1: Neutrino oscillation probability in vacuum at two flavor case as the function of true neutrino energy (in GeV) is depicted here. The upper (lower) panel represents $\nu_\alpha \to \nu_\beta$ appearance ($\nu_\alpha \to \nu_\alpha$ disappearance) probability using Equation 2.30. In the left panel, we observe the oscillation probability for three choices of mixing angle θ *i.e.*, 40°, 45°, and 55° (shown in red, blue, and green lines), respectively, assuming $L = 1300$ km and $\Delta m^2 = +2.522 \times 10^{-3}$ eV^2. In the right panel, we compute the same but for three different choices of Δm^2, *i.e.*, 2.436×10^{-3} eV^2, 2.522×10^{-3} eV^2, and 2.605×10^{-3} eV^2 (shown by red, blue, and green lines), respectively, assuming a given mixing angle $\theta = 45°$.

Δm^2 induces a phase shift toward higher neutrino energies, resulting in a rightward displacement of the oscillation peak.

2.4 Equllibrium state

2.4.1 TOV equation

2.4.2 Tidal Deformability

The discovery of the non-zero value of θ_{13} in the Daya Bay [11, 12] experiment connects the atmospheric neutrino sector and the solar neutrino sector, establishing the 3ν paradigm on a strong footing. At the same time, the two-flavor neutrino oscillation delineates the values of the solar and atmospheric oscillation parameters, and the 3ν paradigm with non-zero θ_{13} gives rise to the possibilities of observing CP violation in the neutrino sector. Now, we discuss the formalism of the three-flavor neutrino oscillation.

Let, a neutrino of flavor α is produced at $x = 0$ with energy E, then the state of the neutrino at $x = 0$ can be expressed as

$$\begin{aligned} |\nu_\alpha(x=0)\rangle &= |\nu_\alpha\rangle \\ &= \sum_{j=1}^{3} (U_{PMNS})^*_{\alpha j} |\nu_j\rangle \, . \end{aligned} \tag{2.32}$$

At the position $x = L$, the neutrino state can be written as,

$$\begin{aligned} |\nu_\alpha(x=L)\rangle &= \sum_{j=1}^{3} e^{ip_jL} (U_{PMNS})^*_{\alpha j} |\nu_j\rangle \\ &= e^{ip_kL} \sum_{j=1}^{3} e^{i(p_j-p_k)L} (U_{PMNS})^*_{\alpha j} |\nu_j\rangle \, . \end{aligned} \tag{2.33}$$

Assuming $m_j << E$ we can approximate,

$$
\begin{aligned}
p_j &= \sqrt{E^2 - m_j^2} \\
&= E - \frac{m_j^2}{2E} + ...
\end{aligned}
\tag{2.34}
$$

.

So,

$$
p_j - p_k \approx -\frac{\Delta m_{jk}^2}{2E} \tag{2.35}
$$

where, $\Delta m_{jk}^2 = m_j^2 - m_k^2$.

Hence, we can write

$$
|\nu_\alpha(x = L)\rangle = e^{ip_k L} \sum_{j=1}^{3} exp\left(-i\frac{\Delta m_{jk}^2\, L}{2E}\right) (U_{PMNS})_{\alpha j}^* \, |\nu_j\rangle \, . \tag{2.36}
$$

So, the probability amplitude of observing the neutrino flavor β at $x = L$ is given by (neglecting the overall phase as it will be cancelled out while multiplying with its complex conjugate while obtaining the probability),

$$
\begin{aligned}
\mathrm{A}_{\beta\alpha} &= \langle \nu_\beta | \nu_\alpha(x = L) \rangle \\
&= \left[\sum_{k=1}^{3} \langle \nu_k | \, (U_{PMNS})_{\beta k} \right] \left[\sum_{j=1}^{3} \exp\left(-i\frac{\Delta m_{jk}^2\, L}{2E}\right) (U_{PMNS})_{\alpha j}^* \, |\nu_j\rangle \right] \\
&= \sum_{j=1}^{3} (U_{PMNS})_{\beta j} \exp\left(-i\frac{\Delta m_{jk}^2\, L}{2E}\right) (U_{PMNS})_{\alpha j}^* \\
&= \sum_{j=1}^{3} U_{\beta j} \exp\left(-i\frac{\Delta m_{jk}^2\, L}{2E}\right) U_{\alpha j}^* .
\end{aligned}
\tag{2.37}
$$

So, the probability of oscillation from $|\nu_\alpha\rangle$ to $|\nu_\beta\rangle$ with true neutrino energy E and baseline L is given by,

$$
\begin{aligned}
P(\nu_\alpha \to \nu_\beta) &= |A_{\beta\alpha}|^2 \\
&= |\sum_{j=1}^{3} U_{\beta j} exp\left(-i\frac{\Delta m_{jk}^2 L}{2E}\right) U_{\alpha j}^* |^2 \\
&= \left[\sum_{j=1}^{3} U_{\beta j} exp\left(-i\frac{\Delta m_{jk}^2 L}{2E}\right) U_{\alpha j}^*\right] \times \left[\sum_{i=1}^{3} U_{\alpha i} exp\left(i\frac{\Delta m_{ik}^2 L}{2E}\right) U_{\beta i}^*\right] \\
&= \sum_{j=1}^{3}\sum_{i=1}^{3} exp\left[-i\frac{\left(\Delta m_{jk}^2 - \Delta m_{ik}^2\right) L}{2E}\right] U_{\beta j} U_{\alpha j}^* U_{\alpha i} U_{\beta i}^* \qquad (2.38)
\end{aligned}
$$

$$
= \sum_i \sum_j (A_{ij} + iB_{ij}) \times (C_{ij} + iD_{ij})\,;(say), \qquad (2.39)
$$

where we assume, $exp\left[-i\frac{\left(\Delta m_{jk}^2 - \Delta m_{ik}^2\right) L}{2E}\right] = A_{ij} + iB_{ij}$ and $U_{\beta j} U_{\alpha j}^* U_{\alpha i} U_{\beta i}^* = C_{ij} + iD_{ij}$. As the oscillation probability is a measurable quantity, so it should be real. Hence, we can drop the imaginary component of equation 2.39.

We break the sum of equation 2.39 in two parts, *i.e.*,

$$
\sum_{i=1}^{3}\sum_{j=1}^{3} = \sum_{i=j=1}^{3} + \sum_{\substack{i\neq j \\ i=1,\, j=1}}^{3} .
$$

<u>Case - 1 :</u>

for $i = j$,

$$
exp\left[-i\frac{\Delta m_{ii}^2 L}{2E}\right] U_{\beta i} U_{\alpha i}^* U_{\alpha i} U_{\beta i}^* = U_{\beta i} U_{\alpha i}^* U_{\alpha i} U_{\beta i}^* . \qquad (2.40)
$$

<u>Case - 2 :</u>

for $i \neq j$,

$$Re\left(exp\left[-i\frac{\Delta m_{ji}^2\, L}{2E}\right]\right) = \cos\left(\frac{\Delta m_{ji}^2\, L}{2E}\right)$$
$$= \cos(\Delta_{ji});\;\; (say, A_{ij}), \tag{2.41}$$
$$Im\left(exp\left[-i\frac{\Delta m_{ji}^2\, L}{2E}\right]\right) = -\sin\left(\frac{\Delta m_{ji}^2\, L}{2E}\right)$$
$$= -\sin(\Delta_{ji});\;\; (say, B_{ij}). \tag{2.42}$$

So, from equation 2.39, considering the real part only, we can write

$$\begin{aligned} P(\nu_\alpha \to \nu_\beta) &= A_{ij}C_{ij} - B_{ij}D_{ij} \\ &= Re(U^*_{\alpha i}U_{\beta i}U_{\alpha j}U^*_{\beta j}) - 2\,Re(U^*_{\alpha i}U_{\beta i}U_{\alpha j}U^*_{\beta j})\sin^2\left(\frac{\Delta_{ji}}{2}\right) \\ &\quad + Im\,(U^*_{\alpha i}U_{\beta i}U_{\alpha j}U^*_{\beta j})\sin(\Delta_{ji}). \end{aligned} \tag{2.43}$$

Here, we have used $\cos(\Delta_{ji}) = 1 - 2\sin^2\left(\frac{\Delta_{ji}}{2}\right)$. So, taking case - 1 (*i.e.*, $i = j$) and case - 2 ($i \neq j$) together, we can write,

$$\begin{aligned} P(\nu_\alpha \to \nu_\beta) = \sum_i \sum_j \Big[& U_{\beta i}U^*_{\alpha i}U_{\alpha i}U^*_{\beta i} + Re\,(U^*_{\alpha i}U_{\beta i}U_{\alpha j}U^*_{\beta j}) \\ & - 2Re\,(U^*_{\alpha i}U_{\beta i}U_{\alpha j}U^*_{\beta j})\sin^2\left(\frac{\Delta_{ji}}{2}\right) \\ & + Im\,(U^*_{\alpha i}U_{\beta i}U_{\alpha j}U^*_{\beta j})\sin(\Delta_{ji})\Big]. \end{aligned} \tag{2.44}$$

Now,

$$\begin{aligned} \left(\sum_{i=1}^{3} U^*_{\alpha i}U_{\beta i}\right)^2 &= (U^*_{\alpha 1}U_{\beta 1} + U^*_{\alpha 2}U_{\beta 2} + U^*_{\alpha 3}U_{\beta 3})^2 \\ &= \sum_i (U^*_{\alpha i}U_{\beta i})^2 + \sum_{i\neq j} 2(U^*_{\alpha i}U_{\beta i}U_{\alpha j}U^*_{\beta j}). \end{aligned} \tag{2.45}$$

So, from the equation 2.45, we can write the final generalized form of the equation of neutrino oscillation probability in 3ν framework as

$$P(\nu_\alpha \to \nu_\beta) = \delta_{\alpha\beta} - 4\sum_{i>j} Re\,(U^*_{\alpha i}U_{\beta i}U_{\alpha j}U^*_{\beta j})\sin^2\left(\frac{\Delta_{ji}}{2}\right) \\ + 2\sum_{i>j} Im\,(U^*_{\alpha i}U_{\beta i}U_{\alpha j}U^*_{\beta j})\sin(\Delta_{ji}). \tag{2.46}$$

If i and j take only values from 1 to 2, the above equation reduces to the two-flavor neutrino oscillation expression. The third term of equation 2.46 represents the CP violation. The PMNS matrix can be written as,

$$U = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix} \begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta} \\ 0 & 1 & 0 \\ -s_{13}e^{i\delta} & 0 & c_{13} \end{pmatrix} \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{13}s_{23}e^{i\delta} & c_{12}c_{23} - s_{12}s_{13}s_{23}e^{i\delta} & c_{13}s_{23} \\ s_{12}s_{23} - c_{12}s_{13}c_{23}e^{i\delta} & -c_{12}s_{23} - s_{12}s_{13}c_{23}e^{i\delta} & c_{13}c_{23} \end{pmatrix} \tag{2.47}$$

$$\equiv \begin{pmatrix} U_{\alpha 1} & U_{\alpha 2} & U_{\alpha 3} \\ U_{\beta 1} & U_{\beta 2} & U_{\beta 3} \\ U_{\gamma 1} & U_{\gamma 2} & U_{\gamma 3} \end{pmatrix}. \tag{2.48}$$

Here, c_{ij} and s_{ij} stand for $\cos\theta_{ij}$ and $\sin\theta_{ij}$, respectively, where $i, j = 1, 2, 3;\ i \neq j$. We are ignoring the Majorana phases here as they do not affect the neutrino oscillation probability. The advantage of considering the PMNS matrix as a suitable unitary mixing matrix connecting the flavor basis and mass basis of neutrinos is that the ratios of some matrix elements of U give us the values of the oscillation parameters distinctly. For example,

$$\tan^2\theta_{23} = \frac{|U_{\beta 3}|^2}{|U_{\gamma 3}|^2},$$

and,

$$\tan^2\theta_{12} = \frac{|U_{\alpha 2}|^2}{|U_{\alpha 1}|^2}.$$

The 1-3 element of the PMNS matrix gives us the value of δ_{CP} if the value of θ_{13} is well known. The ratios of other matrix elements of U also give the value of the oscillation parameters, *e.g.*,

$$\cos^2\theta_{12} = \frac{|U_{\alpha 1}|^2}{1-|U_{\alpha 3}|^2},\ \sin^2\theta_{12} = \frac{|U_{\alpha 2}|^2}{1-|U_{\alpha 3}|^2},\ \cos^2\theta_{23} = \frac{|U_{\gamma 3}|^2}{1-|U_{\alpha 3}|^2},\ \sin^2\theta_{23} = \frac{|U_{\beta 3}|^2}{1-|U_{\alpha 3}|^2}.$$

Now, substituting the values of the matrix elements from equation 2.47 to equation 2.46, we get the final expression of neutrino oscillation probability in vacuum. If we focus on appearance channels only (*i.e.*, $\alpha \neq \beta$ in equation 2.46) and keep the terms upto second order of $(\alpha \cdot \sin(2\theta_{13}))$, where $\alpha = \frac{\Delta m^2_{21}}{\Delta m^2_{31}}$, then equation 2.46 takes the following form [13].

$$\begin{aligned} P(\nu_\mu \to \nu_e) \simeq\ & \boldsymbol{\sin^2\theta_{23}} \sin^2(2\theta_{13}) \sin^2\Delta \\ & + (\alpha\Delta)\sin(2\theta_{13})\sin(2\theta_{12})\sin(2\theta_{23})\, \boldsymbol{\sin(\delta_{\mathrm{CP}})} \sin^2(\Delta) \\ & + (\alpha\Delta)\sin(2\theta_{13})\sin(2\theta_{12})\sin(2\theta_{23})\, \boldsymbol{\cos(\delta_{\mathrm{CP}})} \cos(\Delta)\sin(\Delta) \\ & + (\alpha\Delta)^2\cos^2\theta_{23}\sin^2(2\theta_{12}). \end{aligned} \tag{2.49}$$

The first term in the equation 2.49 is the leading term as it does not depend on α (as $\alpha << 1$) and governed by the atmospheric mixing angle θ_{23}. Here, $\Delta = \frac{\Delta m^2_{31} L}{4E}$. Note that, this term depends on $\sin^2\theta_{23}$ and sensitive to the octant of θ_{23}, hence the $\nu_\mu \to \nu_e$ appearance channel helps to resolve the octant of θ_{23}. Although, there is a sub-leading effect of the second term, it gives the signature of CP violation as $\sin(\delta_{\mathrm{CP}})$ changes sign as δ_{CP} changes its sign when we go from neutrino to antineutrino, *i.e.*, it is a CP-odd term. Hence, the $\nu_\mu \to \nu_e$ appearance channel helps to study CP violation (Dirac $\boldsymbol{\delta_{\mathrm{CP}}}$) in the neutrino sector. The third term is the CP-even term. The last term is the solar term and depends on the solar mixing angle θ_{12}. But, its contribution in the appearance channel is very much suppressed due to its α^2 dependence.

Similarly, we can study some features of the $\nu_\mu \to \nu_\mu$ disappearance probability in vacuum by writing its expression following the previous approximations,

$$\begin{aligned}P(\nu_\mu \to \nu_\mu) \simeq 1 &- \mathbf{\sin^2 2\boldsymbol{\theta}_{23}} \sin^2(\Delta) \\ &+ 4\sin^2\theta_{13}\sin^2\theta_{23}\cos(2\theta_{23})\sin^2\Delta \\ &+ (\alpha\Delta)\sin 2\Delta[\cos^2\theta_{12}\sin^2 2\theta_{23} - 2\sin\theta_{13}\sin 2\theta_{12}\sin^2\theta_{23}\sin 2\theta_{23}\,\mathbf{\cos\boldsymbol{\delta}_{CP}}] \\ &- (\alpha\Delta)^2[\sin^2 2\theta_{12}\cos^2\theta_{23} + \cos^2\theta_{12}\sin^2 2\theta_{23}(\cos 2\Delta - \sin^2\theta_{12})].\end{aligned} \tag{2.50}$$

The first salient feature of the above expression is that the leading term is proportional to $\sin^2 2\theta_{23}$ and therefore it is not sensitive to the octant of θ_{23}, but its strength is larger than the $\nu_\mu \to \nu_e$ appearance probability in equation 2.49, which is suppressed by $\sin^2(2\theta_{13})$. Hence, the disappearance channel is rich in statistics. Furthermore, this channel is not sensitive to the CP violation as it only contains $\cos\delta_{\rm CP}$ term. The motivations behind showcasing only these two channels throughout this thesis are i) the $\nu_\mu \to \nu_e$ appearance channel [14, 15] is sensitive to two of the most important unresolved issues in the 3ν paradigm, *i.e.*, CP violation and exclusion of the wrong octant solution of θ_{23}, ii) the $\nu_\mu \to \nu_\mu$ disappearance channel ensures to bring good precision in measuring the atmospheric oscillation parameters θ_{23} and Δm^2_{31} owing to high statistics in this channel in long-baseline (LBL) experiments (for details, see the next chapter on long-baseline experiments). For antineutrinos, the sign of $\delta_{\rm CP}$ changes its sign (*i.e.*, $P(\nu_\alpha \to \nu_\beta;\, \delta_{\rm CP}) \to P(\bar\nu_\alpha \to \bar\nu_\beta;\, -\delta_{\rm CP})$).

The CP-odd asymmetry in neutrino oscillation can be written as,

$$\begin{aligned}\Delta P_{\alpha\beta} &= P(\nu_\alpha \to \nu_\beta) - P(\bar\nu_\alpha \to \bar\nu_\beta) \\ &= \pm 16 J \sin\left(\frac{\Delta m^2_{21}L}{4E}\right)\sin\left(\frac{\Delta m^2_{31}L}{4E}\right)\sin\left(\frac{\Delta m^2_{32}L}{4E}\right).\end{aligned} \tag{2.51}$$

Here, $\Delta P_{\alpha\beta}$ denotes the amount of CP-odd asymmetry in the neutrino sector, J is the Jarlskog invariant ($J \equiv Im[U_{e1}U^*_{\mu 1}U^*_{e2}U_{\mu 2}]$) [16, 17], which is a constant quantity irrespective of the choice of basis, and can be expressed under the standard PMNS parameterization as,

$$J = \frac{1}{8} \cos\theta_{13} \sin(2\theta_{12}) \sin(2\theta_{13}) \sin(2\theta_{23}) \sin\delta_{\text{CP}}. \tag{2.52}$$

Equations 2.51 and 2.52 denote the necessary conditions for CP violation in the neutrino sector [18], *i.e.*,

- the mixing angles $\neq 0°$, $90°$,
- the value of $\delta_{\text{CP}} \neq 0°, \pm 180°$,
- the value of $\Delta m^2_{ij} \neq 0$.

2.5 Perturbed state

2.5.1 f-mode eigenvalue problem

2.5.2 w-mode eigenvalue problem

Neutrinos interact with the ambient matter via weak interactions and it can modify the oscillation probabilities of neutrinos. The first experimental evidence of matter effects in neutrino oscillations came from the solar neutrino experiments, particularly the Sudbury Neutrino Observatory (SNO) [19] and Super-Kamiokande [20], around the early 2000s. These experiments confirmed the Mikheyev-Smirnov-Wolfenstein (MSW) effect [21–23], which describes how the oscillation probability of neutrinos gets modified as they travel through dense matter.

The modified Hamiltonian incorporating the neutrino-matter interaction can be written as,

$$\hat{H}_{eff} = \hat{H}_{vac} + \hat{H}_{matt}. \tag{2.53}$$

To have a close look at how the eigenvalues are modified under the influence of the neutrino-matter interaction, we take the help of the time-dependent Schrödinger equation, *i.e.*,

$$i \frac{d}{dt} |\nu(t)\rangle = \hat{H}_{eff} |\nu(t)\rangle . \tag{2.54}$$

This equation 2.54 can be rewritten in the flavor eigenstate as,

$$i\frac{d}{dt}\nu_\alpha(t) = \sum_\beta H_{\alpha\beta}\nu_\beta(t), \tag{2.55}$$

where α, β denote the flavor indices and $H_{\alpha\beta} = \langle\nu_\alpha|H|\nu_\beta\rangle$ are the matrix elements. On the other hand, $|\nu_k\rangle$ denotes the mass eigenstate and hence the eigenstate of the free Hamiltonian $\hat{H}_{vac}$, *i.e.*,

$$\hat{H}_{vac}\,|\nu_k\rangle = E_k\,|\nu_k\rangle\,, \tag{2.56}$$

where, E_k denotes the eigenvalue of the free Hamiltonian and can be written as $E_k = \sqrt{\vec{p}^2 + m_k^2}$, $\vec{p}$ being the momentum of the mass eigenstate $|\nu_k\rangle$. Neutrinos interact with the electrons of the ordinary matter through charged-current (CC) interaction via the $W^\pm$ boson of the SM. This interaction induces a matter potential confronting which the neutrino changes its flavor while propagating through a medium. This potential can be depicted as,

$$\begin{aligned} V_{CC,\alpha} &= \sqrt{2}G_F N_e, \quad [\alpha = e] \\ &= 0, \quad [\alpha = \mu, \tau]. \end{aligned} \tag{2.57}$$

Neutrinos mainly interact with the ambient electron of the matter while performing charged current interaction. There is another kind of interaction of neutrinos with the matter via the neutral mediator Z^0 boson of the SM, hence it is called neutral current (NC) interaction [24], the matter potential of which can be expressed as,

$$V_{NC,\alpha} = -\frac{G_F}{\sqrt{2}}N_n, \quad [\alpha = e, \mu, \tau], \tag{2.58}$$

,

where, N_e and N_n signify the electron number and neutron number densities of the matter, and G_F denotes the Fermi coupling constant ($G_F = 1.166 \times 10^{-5}$ GeV^{-2}). Neutrinos interact with the matter via coherent and forward scattering [25]. Due to the scarcity of naturally produced muon and tau in the Earth matter, CC interaction only occurs with the electrons of the medium. Now, the CC interaction potential can be expressed in terms of some measurable parameters used in the experiments directly, *i.e.*, line-averaged Earth matter density ρ_{avg} [26] and the above-mentioned number densities (*i.e.*, N_e and N_n) as the following fashion,

$$V_{CC} \approx 7.6 \times \left(\frac{N_e}{N_p + N_n}\right) \times \left[\frac{\rho_{\text{avg}}}{10^{14}\,\text{g}/\text{cm}^3}\right] \text{eV}. \tag{2.59}$$

Considering further the medium of interaction is electrically neutral (*i.e.*, $N_e = N_p$) and isoscaler (*i.e.*, $N_p = N_n$), equation 2.59 can be rewritten as,

$$V_{CC} = 7.6 \times 0.5 \times \left[\frac{\rho_{\text{avg}}}{10^{14}\,\text{g}/\text{cm}^3}\right] \text{eV}. \tag{2.60}$$

The total matter potential $V(\nu)_\alpha = V_{CC,\alpha} + V_{NC,\alpha}$ changes its polarity while considering antineutrinos instead of neutrinos *i.e.*, $V(\nu)_\alpha = -V(\bar{\nu})_\alpha$. This V_α is the eigenvalue of the Hamiltonian $\hat{H}_{\text{matt}}$ in weak basis, *i.e.*,

$$\hat{H}_{\text{matt}} \left|\nu_\alpha\right\rangle = V_\alpha \left|\nu_\alpha\right\rangle. \tag{2.61}$$

At tree level, neutral-current interactions mediated by the Z^0 boson are flavour universal, giving identical contributions to all active neutrinos, and hence it does not affect the oscillation probability. However, at the one-loop level, radiative corrections to the neutrino–Z^0 vertex involve virtual W bosons and the corresponding charged lepton partner. Since these corrections depend on the charged lepton mass, the large difference between m_μ and m_τ breaks flavor universality, leading to a small but finite splitting between the effective matter potentials of ν_μ and ν_τ [27]. The difference between the matter potential confronted by ν_μ and ν_τ can be written as [28],

$$\Delta V_{\tau\mu} = V_\tau - V_\mu = -\frac{3\sqrt{2}}{2\pi^2} G_F m_\tau^2 \left(ln\frac{m_W}{m_\tau} - \frac{1}{2}\right)\frac{N_e}{m_W^2}, \tag{2.62}$$

where, m_τ and m_W denote mass of tau and W boson. For Earth matter, the relative contribution of this splitting with respect to the potential $V_{\text{CC,e}}$ is

$$\frac{\Delta V_{\tau\mu}}{V_{\text{CC,e}}} \sim -2.5 \times 10^{-4}. \tag{2.63}$$

Although in standard interaction of active neutrinos, this effect is negligible, but from the context of dense environments like supernova, this slight difference between the interaction potentials becomes significant. It is the first non-zero splitting for the NC interaction of ν_μ and ν_τ (or in the 2-3 sector) which could affect neutrino oscillation in dense media.

From equation 2.56, it is evident that the energy of the neutrinos is the eigenvalues of the vacuum Hamiltonian in mass basis. To convert it into the physically measurable flavor basis, we need to diagonalize it through the unitary mixing matrix U by the following equation

$$\hat{H}^f_{vac} = U\hat{H}^m_{vac}U^\dagger. \tag{2.64}$$

. In the matrix form, it can be written as,

$$H^f = \mathbf{U}\begin{bmatrix} E^m_1 & 0 & 0 \\ 0 & E^m_2 & 0 \\ 0 & 0 & E^m_3 \end{bmatrix}\mathbf{U}^\dagger. \tag{2.65}$$

After invoking the influence of the matter effect in our picture, this mixing matrix U cannot diagonalize the effective Hamiltonian $\hat{H}_{eff}$, and it will not be diagonal even on the previous mass basis, considered so far. Hence, we need to construct a new type of mass basis (say, $|\nu^m_i\rangle$) where $\hat{H}_{vac}$ is diagonal. In this matter-modified mass basis $|\nu^m_i\rangle$, the diagonalized form of the effective Hamiltonian will be $\hat{H}^m = diag\,(E^m_1, E^m_2, E^m_3) = diag\,\frac{1}{2E}(\tilde{m}^2_1, \tilde{m}^2_2, \tilde{m}^3_3) = \tilde{U}\hat{H}_{eff}\tilde{U}^\dagger$. Here, $\tilde{U}$ is the matter-modified mixing matrix (kindly note that, $|\nu^m_i\rangle \neq |\nu_i\rangle\,, \tilde{U}^m \neq U$). Like the previous case, here also we assume equal momentum approximation ($i.e., E_1 = E_2 = E_3 = E$) and $\tilde{m}^2_i$ signifies the matter-modified mass squared term. So, the flavor basis can now be expressed in terms of the matter-modified mass basis through this relation $|\nu_\alpha\rangle = \tilde{U}\,|\nu^m_i\rangle$. Adorned with this new mass basis and new mixing matrix, we can show the neutrino oscillation probability while neutrinos interact through matter with the two new plug-ins, $U_{\alpha i} \to \tilde{U}^m_{\alpha i}$ and $E_i \to E^m_i$, where E^m_i is the matter-modified energy eigenvalue of each mass state. Now, following the footprints of the same calculations mentioned in the section 2.4.2, we can write down the expression of the neutrino oscillation probability under the influence of the matter effect as,

$$P(\nu_\alpha \to \nu_\beta) = \delta_{\alpha\beta} - 4\sum_{i>j} Re(\tilde{U}^*_{\alpha i}\tilde{U}_{\beta i}\tilde{U}_{\alpha j}\tilde{U}^*_{\beta j})\sin^2\left(\frac{\tilde{\Delta}_{ji}}{2}\right)$$
$$+ 2\sum_{i>j} Im(\tilde{U}^*_{\alpha i}\tilde{U}_{\beta i}\tilde{U}_{\alpha j}\tilde{U}^*_{\beta j})\sin(\tilde{\Delta}_{ji}), \qquad (2.66)$$

where, $\tilde{\Delta}_{ji} = \dfrac{\Delta\tilde{m}^2_{ji}L}{4E}$ denotes the modified frequency term with two measurable quantities, baseline length (L in km) and the neutrino energy (E in GeV).

CHAPTER 3

Dark matter admixed quarkyonic stars

3.0.1 Introduction

3.0.2 Results and Discussions

3.0.3 Conclusion

CHAPTER 4

f-mode Oscillation of Dark Matter Admixed Quarkyonic Star

4.0.1 Introduction

4.0.2 Results and Discussions

4.0.3 Conclusion

CHAPTER 5

w-mode Oscillation of Dark Matter Admixed Quarkyonic Star

5.0.1 Introduction

5.0.2 Results and Discussions

5.0.3 Conclusion

CHAPTER 6

Universality in Space-Time ω Modes of Quarkyonic Star

6.0.1 Introduction

6.0.2 Results and Discussions

6.0.3 Conclusion

CHAPTER 7

Summary and Conclusions

This thesis addresses some important issues in three-flavor neutrino oscillation in the context of upcoming high-precision long-baseline experiments. Neutrinos are omnipresent and they are the second most abundant particles in the Universe after photons. There are a variety of neutrino sources (both natural and artificial) producing neutrinos over a wide range of energies in the range of 10^{-4} to 10^{18} eV or so and traveling distances from a few meters to hundreds of Gpc. Neutrinos were postulated first by Wolfgang Pauli in 1930 to explain how beta-decay could conserve energy, momentum, and angular momentum. In 1934, Enrico Fermi named the new particle as "neutrino", which means the "little neutral one" in the Italian language. In 1956, Clyde L. Cowan and Frederick Reines first observed neutrinos in a nuclear reactor. There are three (3) light active flavor neutrinos: ν_e, ν_μ, and ν_τ - experimentally confirmed by the precision data on the Z-decay width at the e^+e^- collider at LEP [33]. They only take part in weak interactions through massive $W^\pm$ and Z^0 gauge bosons. They are electrically neutral, left-handed, and form $SU(2)$ doublets in the Standard Model with the corresponding left-handed charged leptons. In the basic SM of particle physics, neutrinos are massless due to the absence of right-handed neutrinos.

Over the past two decades, data from several world-class neutrino experiments firmly established that neutrinos change flavor after propagating a finite distance in space and time - a phenomenon known as "neutrino oscillation". Neutrino oscillation demands non-zero neutrino mass and mixing. Hence, non-zero neutrino mass is the first experimental proof for physics beyond the Standard Model for which the Nobel Prize was awarded to Takaaki Kajita and Arthur McDonald in 2015. Neutrino oscillation, in the 3ν paradigm, is mainly governed by six oscillation parameters, viz., solar mixing angle (θ_{12}), atmospheric mixing angle (θ_{23}), reactor mixing angle (θ_{13}), Dirac CP-violating phase ($\delta_{\rm CP}$), solar mass-squared splitting ($\Delta m^2_{21} \equiv m^2_2 - m^2_1$), and atmospheric mass-squared splitting ($\Delta m^2_{31} \equiv m^2_3 - m^2_1$). Still, there are a few fundamental issues that need to be resolved in neutrino oscillation experiments, namely, i) the neutrino mass ordering (whether $\Delta m^2_{31} > 0$, known as normal mass ordering - NMO, or $\Delta m^2_{31} < 0$, termed as inverted mass ordering - IMO), ii) the octant of θ_{23} (whether $\theta_{23} < 45^\circ$ known as lower octant - LO, or $\theta_{23} > 45^\circ$ termed as higher octant - HO), and iii) whether the CP is violated in the neutrino sector like quark sector and if so, then what is the precise value of the CP phase? The next-generation high-precision long-baseline (LBL) experiments are capable to address these issues at a high confidence level.

In this thesis, we study i) the physics reach of the upcoming LBL experiment, Deep Underground Neutrino Experiment (DUNE) in the US, to establish the possible deviation from maximal mixing of θ_{23}, ii) to achieve an improved precision on the 2-3 oscillation parameters using the synergy between the DUNE and Tokai-to-Hyper-Kamiokande (T2HK) experiments, and iii) the possible impact of the flavor-dependent long-range neutrino interactions on the measurements of neutrino oscillation parameters in DUNE and T2HK.

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Appendix

APPENDIX A

Phase Amplitude Method

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